

Ionic Bonding 離子鍵

Electron Transfer · Ion Formation · Dot-and-Cross Diagrams · Properties of Ionic Compounds

Section A — Multiple Choice 選擇題

20 marks

Choose the best answer. Write the letter on the line. (2 marks each)

1. Which of the following best describes an ionic bond?

- A. Sharing of electrons between two non-metal atoms
B. Electrostatic attraction between oppositely charged ions
C. Transfer of protons between atoms
D. Attraction between two positively charged ions

Answer: _____

2. When a sodium atom (Na) forms an ionic bond with chlorine (Cl), what happens?

- A. Na gains one electron; Cl loses one electron
B. Na loses one electron; Cl gains one electron
C. Both atoms share one electron
D. Na gains two electrons; Cl gains one electron

Answer: _____

3. The electron configuration of Na is 2,8,1. What is the electron configuration of Na^+ ?

- A. 2,8,2
B. 2,8,1
C. 2,8
D. 2,9

Answer: _____

4. Which pair of elements is most likely to form an ionic compound?

- A. Carbon and Oxygen
B. Hydrogen and Chlorine
C. Potassium and Bromine
D. Nitrogen and Hydrogen

Answer: _____

5. Magnesium (Mg, 2,8,2) reacts with oxygen (O, 2,6) to form MgO. What is the charge on the magnesium ion?

- A. 1+
B. 2+
C. 2-
D. 1-

Answer: _____

6. Which of the following is NOT a property of ionic compounds?

- A. They have high melting points
B. They conduct electricity when dissolved in water
C. They are always gases at room temperature
D. They form a giant ionic lattice structure

Answer: _____

7. Calcium (Ca) has atomic number 20. When it forms an ion, it becomes:

- A. Ca^+ (loses 1 electron)
B. Ca^{2+} (loses 2 electrons)
C. Ca^{2-} (gains 2 electrons)
D. Ca^- (gains 1 electron)

Answer: _____

8. Why does solid sodium chloride (NaCl) NOT conduct electricity?

- A. It has no ions
B. The ions are fixed in position in the lattice and cannot move
C. It has no electrons
D. The ions cancel each other out

Answer: _____

9. Aluminium (Al) has electron configuration 2,8,3. What ion does it form?

- A. Al^+
B. Al^{2+}
C. Al^{3+}
D. Al^{3-}

Answer: _____

10. The formula of calcium chloride is CaCl_2 . This is because:

- A. Ca^{2+} needs two Cl^- to balance the charges
B. Calcium has 2 atoms
C. Chlorine always forms double bonds
D. Calcium is in Period 2

Answer: _____

Section B — Fill in the Blanks 填充題

20 marks

Word Bank: transfer · electrostatic · gain · lose · noble gas · metal · non-metal · positive · negative · lattice · high · conduct · soluble · electron · ions

1. Ionic bonds are formed by the _____ of electrons from a _____ atom to a _____ atom.
2. After losing electrons, metal atoms become _____ ly charged ions (cations). After gaining electrons, non-metal atoms become _____ ly charged ions (anions).
3. Both ions formed have the electron configuration of a _____.
4. An ionic bond is the _____ attraction between oppositely charged _____.
5. Ionic compounds have _____ melting points because a large amount of energy is needed to break the strong _____ attractions in the giant ionic _____.
6. Ionic compounds _____ electricity when molten or dissolved in water because the ions are free to move and carry charge. In solid state, they do _____ conduct electricity.
7. Metal atoms tend to _____ electrons to achieve a stable outer shell, while non-metal atoms tend to _____ electrons.

Section C — Ion Formation Table 離子形成表格

20 marks

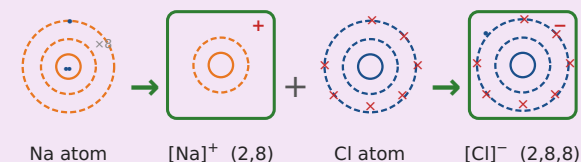
Complete the table. (1 mark per correct cell)

Element	Symbol	Proton No.	e ⁻ Config.	Group	e ⁻ lost/gained	Ion formed	Ion config.
Sodium	Na	11	2,8,1	I	Loses 1		
Magnesium	Mg	12	2,8,2			Mg ²⁺	
Aluminium	Al	13		III	Loses 3		2,8
Chlorine	Cl	17	2,8,7			Cl ⁻	
Oxygen	O	8		VI	Gains 2		2,8
Potassium	K	19	2,8,8,1				2,8,8
Calcium	Ca	20		II		Ca ²⁺	2,8,8
Fluorine	F	9	2,7		Gains 1		
Sulphur	S	16		VI		S ²⁻	2,8,8
Lithium	Li	3	2,1				2

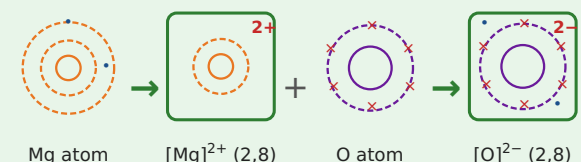
★ How to draw dot-and-cross diagrams for ionic compounds:

- ① Draw the metal atom using **dots (•)** for electrons. Draw the non-metal atom using **crosses (x)** for electrons.
- ② Show the transfer of electron(s) with an arrow (→).
- ③ Draw the resulting ions in **square brackets []** with the charge shown outside (e.g. $[\text{Na}]^+$).
- ④ Only show the **outer shell** electrons (unless the question asks for all shells).
- ⑤ Both ions should have a **full outer shell** (noble gas configuration).

Part 1: Study these completed examples, then answer the questions below.

Example 1: Sodium Chloride (NaCl) — Na (2,8,1) + Cl (2,8,7)

Na loses 1 electron $\rightarrow \text{Na}^+$ (2,8) | Cl gains 1 electron $\rightarrow \text{Cl}^-$ (2,8,8) | Formula: **NaCl**

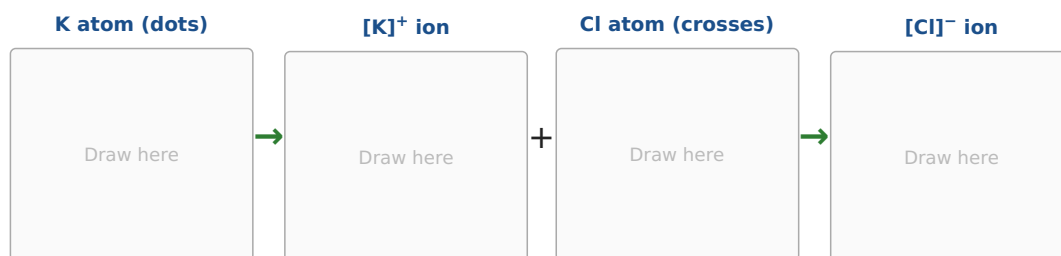
Example 2: Magnesium Oxide (MgO) — Mg (2,8,2) + O (2,6)

Mg loses 2 electrons $\rightarrow \text{Mg}^{2+}$ (2,8) | O gains 2 electrons $\rightarrow \text{O}^{2-}$ (2,8) | Formula: **MgO**

Part 2: Draw dot-and-cross diagrams for the following. Show the transfer of electrons and the ions formed in brackets with charges. (5 marks each)

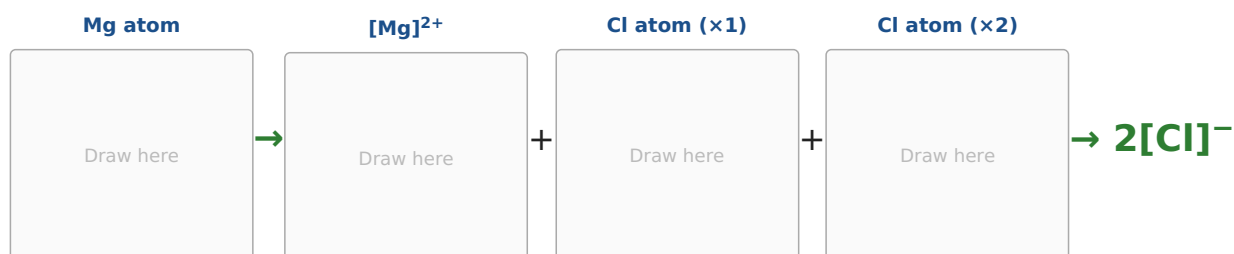
Q1. Potassium Chloride (KCl) — K (2,8,8,1) reacts with Cl (2,8,7)

State: K loses _____ electron(s) $\rightarrow \text{K}$ _____ | Cl gains _____ electron(s) $\rightarrow \text{Cl}$ _____

**Q2. Magnesium Chloride (MgCl₂) — Mg (2,8,2) reacts with 2 Cl (2,8,7)**

Note: Mg loses 2 electrons, so it needs 2 Cl atoms (each gains 1 electron).

State: Mg $\rightarrow$ _____ | Each Cl $\rightarrow$ _____ | Formula: _____



Section E — Ionic Formulae 離子式

10 marks

Rule: The total positive charge must equal the total negative charge (overall charge = 0).

e.g. $\text{Ca}^{2+} + 2\text{Cl}^- \rightarrow \text{CaCl}_2$ ($2+$ balanced by $2 \times 1- = 2-$)

Work out the formula of the ionic compound formed from each pair of ions. Show your working. (1 mark each)

1. Na^+ and O^{2-} Working: _____ Formula: _____
2. Mg^{2+} and Cl^- Working: _____ Formula: _____
3. Al^{3+} and O^{2-} Working: _____ Formula: _____
4. K^+ and S^{2-} Working: _____ Formula: _____
5. Ca^{2+} and N^{3-} Working: _____ Formula: _____
6. Fe^{3+} and Cl^- Working: _____ Formula: _____
7. Li^+ and N^{3-} Working: _____ Formula: _____
8. Ba^{2+} and F^- Working: _____ Formula: _____
9. Al^{3+} and S^{2-} Working: _____ Formula: _____
10. Mg^{2+} and N^{3-} Working: _____ Formula: _____

Section F — Properties & Short Answer 特性及問答

10 marks

Complete the table of properties of ionic compounds. (5 marks)

Property	Observation / Description	Explanation (in terms of ions / structure)
Melting point		
Electrical conductivity (solid)	Does NOT conduct	
Electrical conductivity (molten / dissolved)		Ions are free to move and carry charge
Solubility in water	Most ionic compounds are soluble	
State at room temperature		Strong electrostatic forces hold ions in fixed positions

Short answer questions. (5 marks)

6. Explain why ionic compounds have high melting points. Use the terms "**electrostatic attraction**" and "**giant ionic lattice**" in your answer. (2 marks)

7. Sodium chloride conducts electricity when dissolved in water but not in solid form. Explain this difference. (2 marks)

8. A student says: "When Na becomes Na^+ , it becomes a different element because it has a different number of electrons." Is the student correct? Explain. (1 mark)

ANSWER KEY — For Teacher Use Only

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Section A — MCQ

- | | | | | |
|-------------|-------------|-------------------|-------------|------------------|
| 1. B | 2. B | 3. C (2,8) | 4. C | 5. B (2+) |
| 6. C | 7. B | 8. B | 9. C | 10. A |

Section B — Fill in the Blanks

- | | | | |
|-----------------------------------|------------------------|----------------|-------------------------|
| 1. transfer / metal / non-metal | 2. positive / negative | 3. noble gas | 4. electrostatic / ions |
| 5. high / electrostatic / lattice | 6. conduct / NOT | 7. lose / gain | |

Section C — Ion Formation Table

Element	e ⁻ Config.	Group	e ⁻ lost/gained	Ion	Ion config.
Na	2,8,1	I	Loses 1	Na ⁺	2,8
Mg	2,8,2	II	Loses 2	Mg ²⁺	2,8
Al	2,8,3	III	Loses 3	Al ³⁺	2,8
Cl	2,8,7	VII	Gains 1	Cl ⁻	2,8,8
O	2,6	VI	Gains 2	O ²⁻	2,8
K	2,8,8,1	I	Loses 1	K ⁺	2,8,8
Ca	2,8,8,2	II	Loses 2	Ca ²⁺	2,8,8
F	2,7	VII	Gains 1	F ⁻	2,8
S	2,8,6	VI	Gains 2	S ²⁻	2,8,8
Li	2,1	I	Loses 1	Li ⁺	2

Section D — Dot-and-Cross Diagrams

Q1 KCl: K loses 1 e⁻ → K⁺ (2,8,8) | Cl gains 1 e⁻ → Cl⁻ (2,8,8). Draw K outer shell with 1 dot; K⁺ in brackets, empty outer shell; Cl with 7 crosses; Cl⁻ in brackets with 8 crosses+dot = full outer shell. Award marks for: correct shell diagram (1), correct bracket + charge notation (1), electron transfer shown (1), correct ion config (1), formula stated KCl (1).

Q2 MgCl₂: Mg loses 2 e⁻ → Mg²⁺ (2,8) | Each Cl gains 1 e⁻ → Cl⁻ (2,8,8). Two separate Cl atoms shown each receiving 1 electron (shown as dot in Cl⁻). Formula = MgCl₂. Same marking criteria × 5.

Section E — Formulae

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|---|---|---|---|--|
| 1. 2Na ⁺ :O ²⁻ → Na₂O | 2. Mg ²⁺ :2Cl ⁻ → MgCl₂ | 3. 2Al ³⁺ :3O ²⁻ → Al₂O₃ | 4. 2K ⁺ :S ²⁻ → K₂S | 5. 3Ca ²⁺ :2N ³⁻ → Ca₃N₂ |
| 6. Fe ³⁺ :3Cl ⁻ → FeCl₃ | 7. 3Li ⁺ :N ³⁻ → Li₃N | 8. Ba ²⁺ :2F ⁻ → BaF₂ | 9. 2Al ³⁺ :3S ²⁻ → Al₂S₃ | 10. 3Mg ²⁺ :2N ³⁻ → Mg₃N₂ |

Section F — Properties & Short Answers

Properties Table:

Property	Observation	Explanation
Melting point	High	Strong electrostatic attractions between ions in the giant lattice require large amounts of energy to break
Conductivity (solid)	Does NOT conduct	Ions are fixed in the lattice and cannot move to carry charge
Conductivity (molten/dissolved)	Conducts electricity	Ions are free to move and carry charge
Solubility	Most ionic compounds are soluble	Water molecules attract and surround the ions, pulling them away from the lattice

State at room temp.	Solid	Strong electrostatic forces hold ions in fixed positions
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6. Ionic compounds have high melting points because the ions are held together by **strong electrostatic attractions** in a **giant ionic lattice**. A large amount of energy is needed to overcome these forces and separate the ions.

7. In solid NaCl, the ions are fixed in the lattice and cannot move, so it cannot conduct electricity. When dissolved in water (or molten), the ions are free to move and can carry electrical charge, so it conducts electricity.

8. No. An element is defined by its **number of protons (atomic number)**, not the number of electrons. Na^+ still has 11 protons, so it is still sodium — just a sodium ion with a positive charge.